

SAKAI, Japan

WMO: 477420

Lat: 35.5447N

Lon: 133.2350E

Elev: 12

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.5	32.2	19.5	14.6	38.2	21.7	16.3	39.0	17.8	38.1	16.2	38.6	5.7	250	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	92.4	78.1	90.4	77.7	88.3	77.1	79.7	88.2	78.9	87.1	78.1	85.8	8.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	142.8	83.6	76.6	138.7	83.2	75.8	134.8	82.8	43.3	88.5	42.4	87.2	41.6	86.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.5	11.9	DB	26.4	96.5	2.9	1.8	24.2	97.8	22.5	98.8	20.9	99.8	18.7	101.1	(4)
(5)			WB	24.7	81.1	2.8	1.0	22.7	81.8	21.0	82.3	19.5	82.9	17.4	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	40.7	41.5	47.1	56.1	65.1	71.7	79.6	82.1	74.3	64.5	54.5	45.3	(6)
(7)		DBStd	14.95	4.19	4.78	5.55	5.84	4.70	4.24	4.67	3.93	4.76	4.76	5.24	4.89	(7)
(8)		HDD50	844	290	242	125	13	0	0	0	0	0	0	16	158	(8)
(9)		HDD65	3292	754	657	555	273	57	1	0	1	66	317	611		(9)
(10)		CDD50	4606	1	4	35	195	467	651	919	995	728	449	150	12	(10)
(11)		CDD65	1579	0	0	0	5	59	202	454	530	278	50	1	0	(11)
(12)		CDH74	12930	0	0	1	40	332	933	4192	5638	1654	138	2	0	(12)
(13)		CDH80	4356	0	0	0	3	43	179	1523	2207	382	19	0	0	(13)
(14)	Wind	WSAvg	4.8	5.3	5.1	5.3	5.2	4.9	4.4	4.8	4.6	4.6	4.3	4.5	5.1	(14)
(15)	Precipitation	PrecAvg	72.4	6.9	4.9	5.2	4.4	5.3	6.5	8.8	5.3	8.0	5.0	5.7	6.6	(15)
(16)		PrecMax	91.3	10.4	7.1	7.9	7.5	13.9	14.3	23.2	12.4	16.5	10.8	9.0	10.9	(16)
(17)		PrecMin	60.4	4.2	3.1	2.4	1.5	2.0	1.6	0.4	1.1	2.0	2.0	1.8	2.5	(17)
(18)		PrecStd	8.9	1.9	1.3	1.4	1.8	2.8	3.5	5.9	3.0	4.3	2.6	2.0	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	64.0	70.4	78.7	84.5	88.2	94.2	96.1	91.0	82.3	73.1	62.2	(19)
MCWB			50.8	55.0	55.4	62.0	65.2	73.7	78.1	78.4	77.1	70.9	62.1	54.6	(20)	
2%		DB	52.8	56.5	64.7	73.9	80.4	84.5	91.7	93.5	87.6	77.5	68.6	58.5	(21)	
		MCWB	46.2	49.4	53.0	59.1	63.6	72.0	77.9	78.3	76.2	68.0	60.1	51.5	(22)	
5%		DB	50.2	52.9	60.6	70.4	77.7	81.7	89.8	91.4	84.3	74.6	65.9	56.0	(23)	
		MCWB	44.3	46.9	51.9	57.9	62.9	70.7	77.5	78.1	74.7	66.0	58.4	49.5	(24)	
10%		DB	48.0	50.1	57.2	67.0	74.9	79.2	87.7	89.3	81.5	72.4	63.6	53.7	(25)	
		MCWB	42.6	44.6	50.1	56.5	62.2	69.8	77.1	77.9	73.4	64.5	57.1	48.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	56.0	58.3	66.2	70.2	76.9	80.5	81.0	79.1	73.5	65.1	57.4	(27)
MCD B			56.1	62.1	65.5	71.8	76.7	84.6	89.2	89.7	86.8	79.0	69.0	60.0	(28)	
2%		WB	47.8	50.9	55.4	62.5	67.9	74.9	79.3	80.1	77.4	70.6	62.5	53.1	(29)	
		MCD B	51.6	54.4	61.1	70.7	74.1	80.7	87.8	88.5	84.2	75.5	66.5	56.9	(30)	
5%		WB	45.2	47.9	53.0	59.9	66.0	73.3	78.5	79.3	76.2	68.2	60.1	50.6	(31)	
		MCD B	48.9	52.0	58.5	67.3	73.3	78.5	86.8	87.4	82.3	72.5	64.5	54.7	(32)	
10%		WB	43.2	45.1	50.8	58.1	64.4	71.6	77.7	78.6	75.0	65.9	57.8	48.4	(33)	
		MCD B	47.4	49.4	56.7	64.6	72.0	76.9	85.5	86.5	80.5	70.4	62.6	52.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	12.1	14.7	16.4	15.8	12.2	11.4	12.0	11.3	13.2	13.3	11.5	(35)
MCDBR			15.4	19.2	22.1	22.9	21.6	16.3	15.0	14.9	14.5	16.2	17.5	16.3	(36)	
MCWBR			10.4	13.1	13.4	10.9	9.2	6.7	5.3	5.2	6.0	8.1	10.1	10.6	(37)	
5% WB		MCD BR	14.3	16.5	18.4	17.7	16.0	12.1	13.8	13.2	12.4	12.4	14.3	14.7	(38)	
		MCWBR	10.9	12.9	12.8	10.0	8.8	5.9	5.3	5.1	5.5	7.3	9.8	11.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.438	0.472	0.540	0.557	0.554	0.572	0.569	0.531	0.490	0.453	0.435	0.430	(40)	
taud		2.001	1.943	1.810	1.823	1.888	1.923	1.991	2.099	2.164	2.156	2.108	2.044	(41)		
Ebn at Noon		227	236	232	237	239	235	234	241	244	241	229	221	(42)		
Edh at Noon		43	51	64	66	63	61	56	50	44	41	39	39	(43)		
(44)	All-Sky Solar Radiation	RadAvg	535	771	1140	1509	1666	1562	1541	1603	1241	1010	685	487	(44)	
(45)		RadStd	43	77	102	125	160	120	229	180	107	89	56	41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page